

“It must not be imagined that the secret of the letters can be uncovered with the aid of logical reasoning; it can be reached only through vision and with the aid of divine interpretation.”

ATTRIBUTED TO AHMAD IBN ALI AL-BUNI, 12TH–13TH CENTURY

of the word or words could be found by adding up the numbers for all the letters. This was thought to unlock hidden meanings and enable predictions to be made.

The aim of calculation by nine was to predict winners and losers in a contest or battle by working out the value of each name, dividing by nine, then looking the resulting numbers up on a chart. Similar techniques were used to try to learn the outcome of an illness or journey, or the chance of an event happening. *Jafr* was another technique, which combined the letters in one of the 99 names of God with the letters of a desired objective to help make it happen.

Writing itself was thought to work powerful magic, as well as being a divination tool. Using the right letters was said to give the power to control jinns. There were many treatises devoted to magic alphabets, secret writing, and the letters of ancient cultures, notably the 10th-century scholar Ibn Wahshiyya's *Kitab shawq al-mustaham fi ma'rifat rumuz al-aqlan* (“The book of the frenzied devotee's desire to learn about the riddles of the ancient scripts”). Ibn Wahshiyya was one of the first historians to begin to decipher ancient Egyptian hieroglyphs.

Magic squares

One aspect of Islamic magic that has attracted continual attention is *wafq*, the magic number square. The square may have come from China, but it was adopted in the Arab world, especially

after the 12th century, and appeared in many Islamic magical manuals. The simplest and earliest magic square was the 3x3 *buduh*, in which the numerals 1 to 9 were arranged so that every row and column and both diagonals each added up to 15.

The idea still fascinates mathematicians, who have often tried to create bigger squares. But in the early Islamic world, the interest was in the squares' magical properties, and their ability to ward off afflictions. Indeed, the 3x3 magic square was considered so powerful that simply writing or uttering the name *buduh* might be enough to cure stomach pains or render its user invisible.



▲ Charmed numbers

At the heart of this oval metal seal lies *buduh*, the magic square, with its nine numbers arranged to add up to 15 in any direction. Around the outside are the names of the four archangels.



IN CONTEXT

Arabic alchemy

The aim of alchemy, *al-kimiya* in Arabic, was to turn one substance into another. For 9th-century scholar al-Razi, no one was a true philosopher until they had done this—it compared with the creative power of God. The ultimate quest was to turn base metal into gold, giving eternal life. Alchemists were secretive, in case their techniques fell into the hands of those seeking riches not wisdom. Often later dismissed as charlatans, alchemists laid the basis for chemistry. The greatest, Jabir ibn Hayyan (sometimes known as Geber), gave the world the laboratory, distillation, and strong acids.

An alchemical sage is pictured in a book on dreams by 10th-century alchemist al-Tamimi.